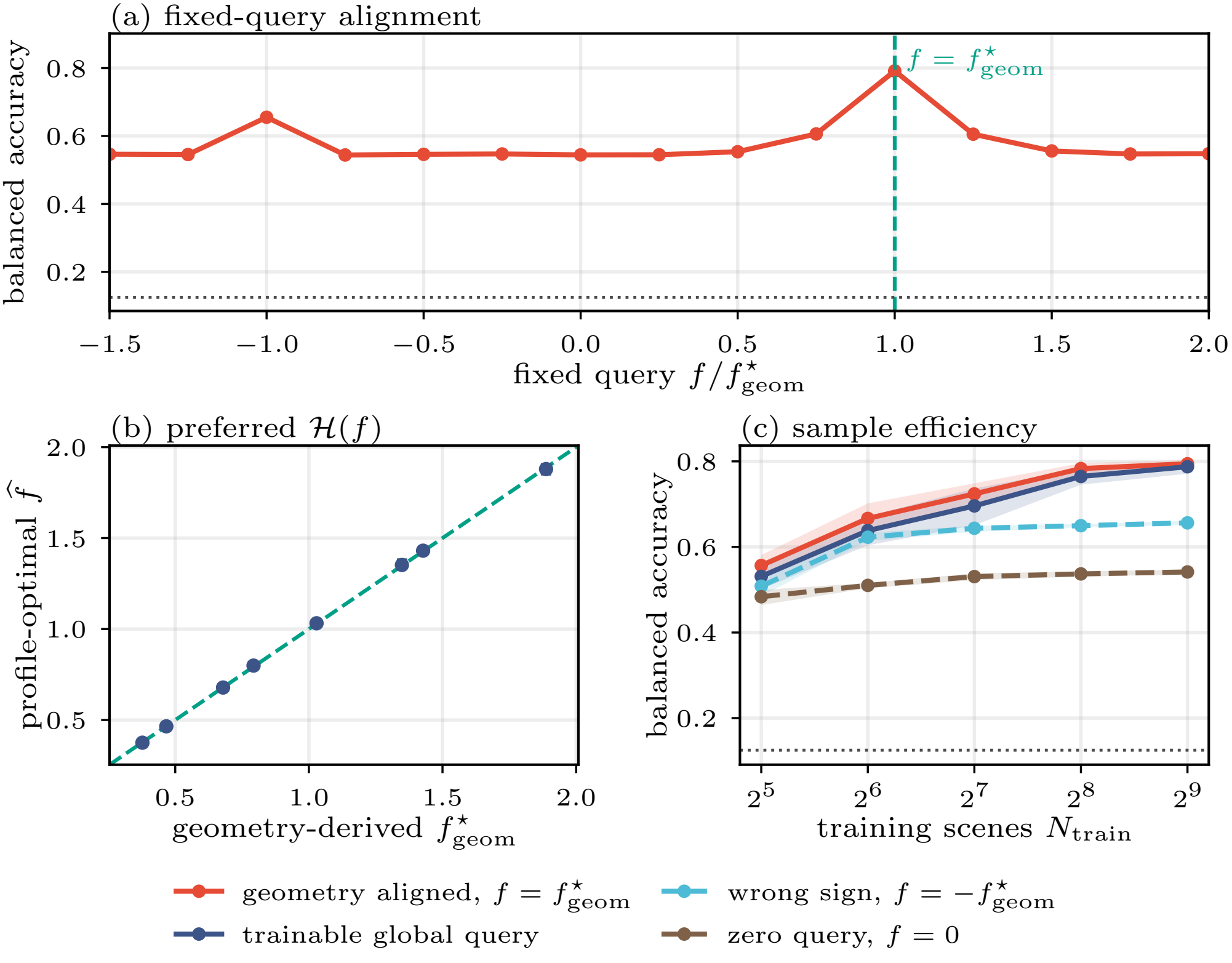

(a) fixed-query alignment
balanced accuracy
0.8
0.6
0.4
0.2
$f = f^{\star}_{\mathrm{geom}}$
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
fixed query $f / f^{\star}_{\mathrm{geom}}$
(b) preferred $\mathcal{H}(f)$
profile-optimal $\widehat{f}$
2.0
1.5
1.0
0.5
0.5
1.0
1.5
2.0
geometry-derived $f^{\star}_{\mathrm{geom}}$
(c) sample efficiency
balanced accuracy
0.8
0.6
0.4
0.2
2^5
2^6
2^7
2^8
2^9
training scenes N_{train}
geometry aligned, $f = f^{\star}_{\mathrm{geom}}$
wrong sign, $f = -f^{\star}_{\mathrm{geom}}$
trainable global query
zero query, $f = 0$